Introduction

Optimization : The procedure or procedures used to make a system or design as effective or functional as possible.

Why Optimization?

- Helps improve the quality of decision-making
- Applications in Engineering, Business, Economics,
- Science, Military Planning etc.

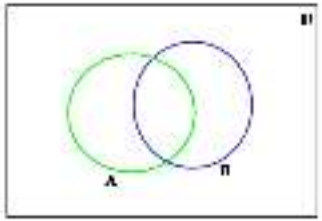
Quite useful for many applications spanning a large number of fields
Design (automotive, aerospace, biomechanical)
Control, Signal processing, Communications, Circuit design

Acknowledgement: 1. NPTEL Course on Numerical Optimization by Prof. Shirish Shevade, IISc Bangalore.
2. Introduction to optimum design 3rd edn. by Jasbir S Arora, Academic Press.

Sets

Definition

A *set* is a collection of objects satisfying certain property P.



Examples:

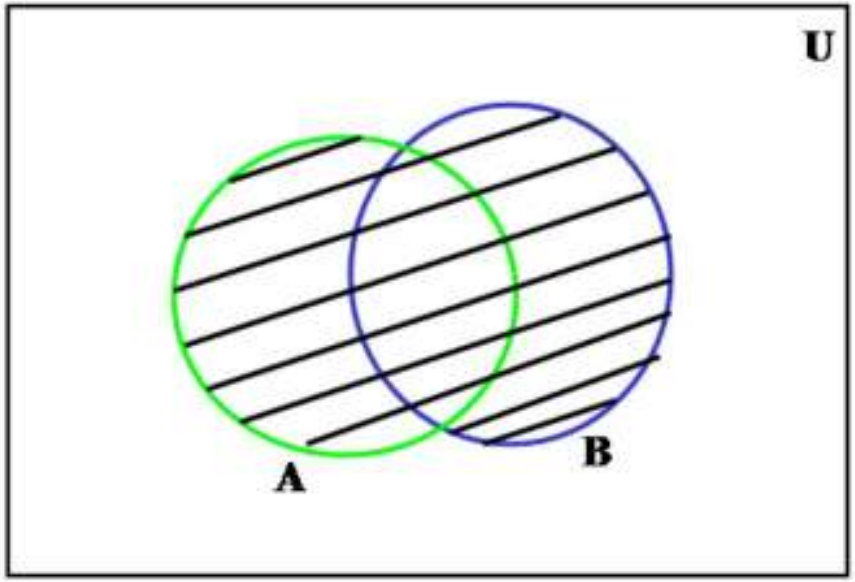
- A set of natural numbers, $\{1, 2, 3, \ldots\}$
- $\{x \in \mathbb{R} : 1 \leq x \leq 3\}$

Note: A set not containing any object is called the *empty* set and is denoted by ϕ.

Let A and B be two sets.

- Union: $A \cup B = \{x : x \in A \text{ or } x \in B\}$
- Intersection: $A \cap B = \{x : x \in A \text{ and } x \in B\}$
- Difference: $A \backslash B = \{x : x \in A \text{ and } x \notin B\}$

Reference :NPTEL Course on Numerical Optimization by Prof. Shirish Shevade, IISc Bangalore.



Union: $A \cup B = \{x : x \in A \text{ or } x \in B\}$

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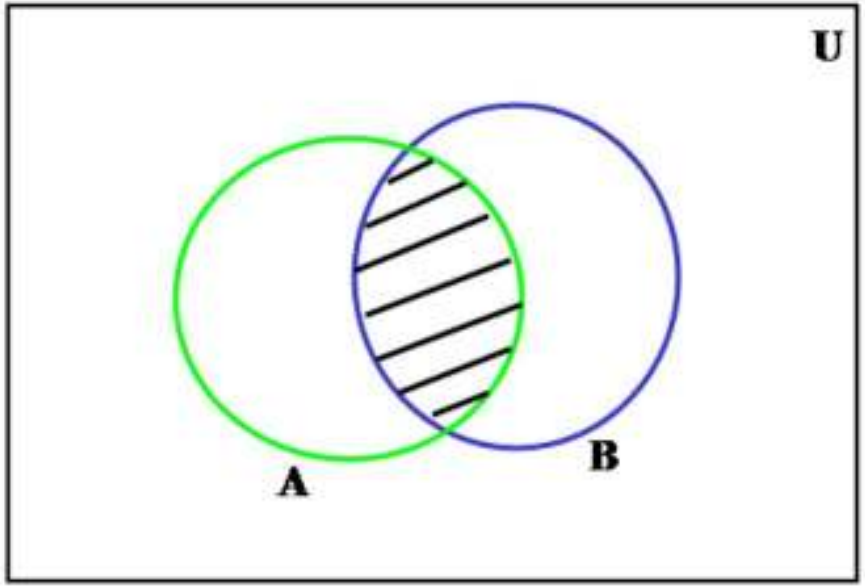

Intersection: $A \cap B = \{x : x \in A \text{ and } x \in B\}$

If the intersection of two sets is empty, we say that the sets are *disjoint*. That is, for two disjoint sets A and B, $A \cap B = \phi$.

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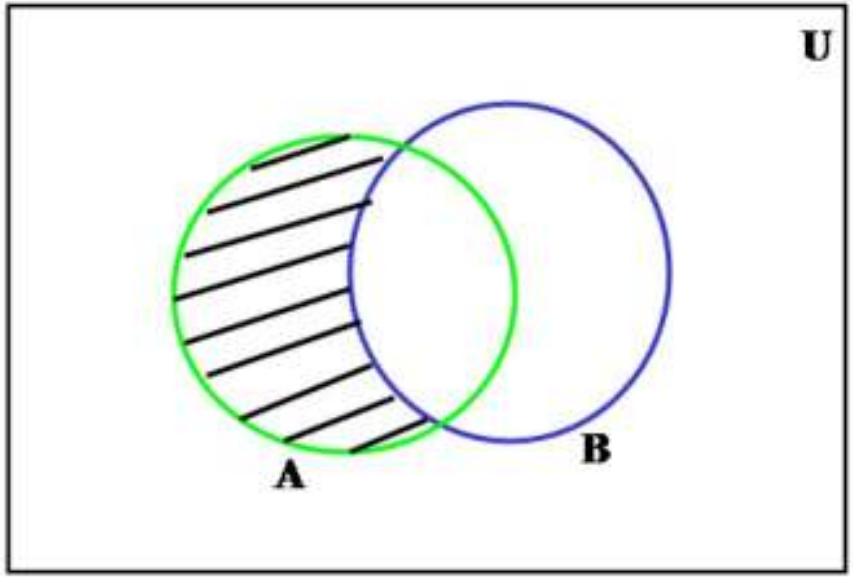

Difference: $A \backslash B = \{x : x \in A \text{ and } x \notin B\}$

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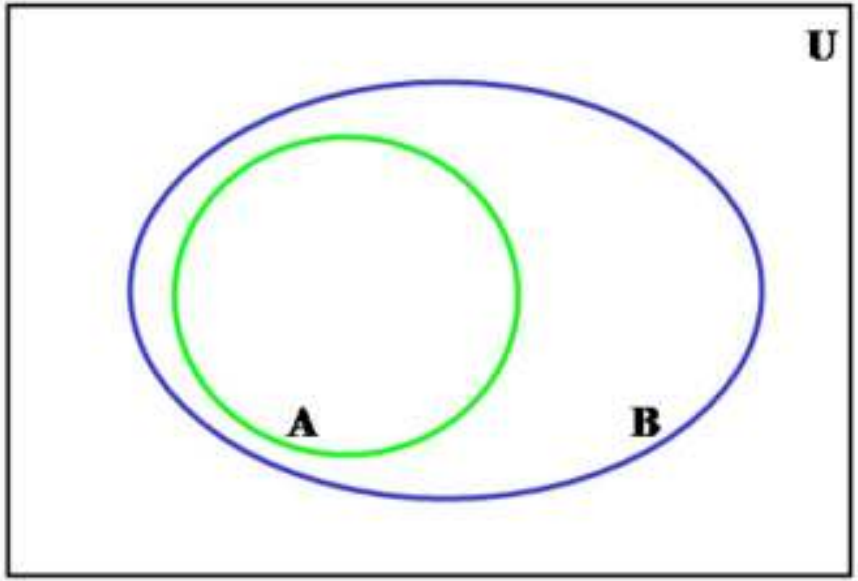

Let A and B be two sets. If A is a subset of B, that is, every member of A is also a member of B, we write $A \subseteq B$. Further, if A is a subset of B and there exists $y \in B$ such that $y \notin A$, then we write $A \subset B$.

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Notations

- $\mathbb{R}$: Vector space of real numbers
- $\mathbb{R}^n$: Vector space of real $n \times 1$ vectors
- n-vector $\mathbf{x}$ is an array of n scalars, $x_1, x_2, \ldots, x_n$
- $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$
- $\mathbf{x} \in \mathbb{R}^n, x_i \in \mathbb{R} \; \forall \; i$
- $\mathbf{x}^T = (x_1, x_2, \ldots, x_n)$
- $\mathbf{0}^T = (0, 0, \ldots, 0)$
- $\mathbf{1}^T = (1, 1, \ldots, 1)$ (We also use $\mathbf{e}$ to denote this vector)

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Functions

Definition

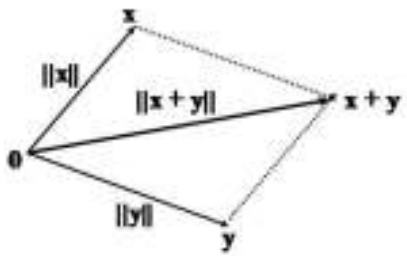
A function f from a set A to a set B is a rule that assigns to each x in A a unique element $f(x)$ in B. This function can be represented by

$$f : A \to B.$$

Definition

A *norm* on $\mathbb{R}^n$ is a real-valued function $\|\cdot\| : \mathbb{R}^n \to \mathbb{R}$ which obeys

- $\|\mathbf{x}\| \geq 0$ for every $\mathbf{x} \in \mathbb{R}^n$, and $\|\mathbf{x}\| = 0$ if and only if $\mathbf{x} = \mathbf{0}$,
- $\|\alpha\mathbf{x}\| = |\alpha|\|\mathbf{x}\|$ for every $\mathbf{x} \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$, and
- $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$ for every $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{y} \in \mathbb{R}^n$.



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Let $\mathbf{x} \in \mathbb{R}^n$.

Some popular norms:

- L_2 or Euclidean norm

$$\|\mathbf{x}\|_2 = \left(\sum_{i=1}^{n} (x_i)^2 \right)^{\frac{1}{2}}$$

- L_1 norm

$$\|\mathbf{x}\|_1 = \sum_{i=1}^{n} |x_i|$$

- L_∞ norm

$$\|\mathbf{x}\|_\infty = \max_{i=1,\ldots,n} |x_i|$$

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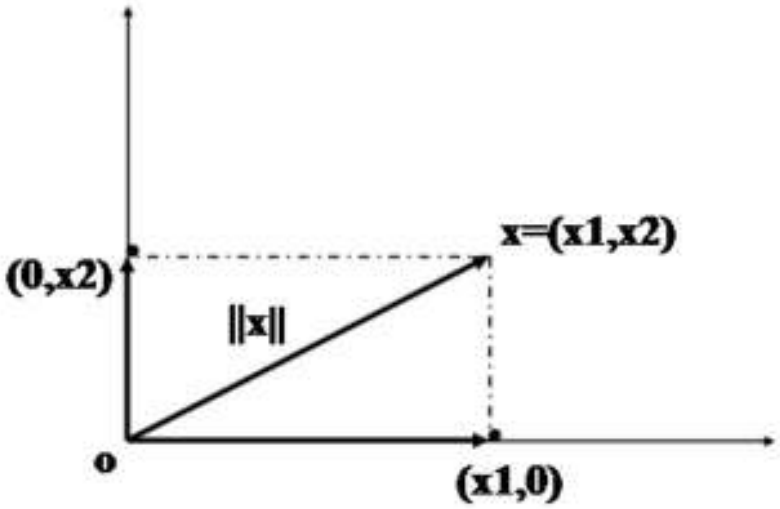
Illustration of L_2 norm:

$$\|\mathbf{x}\|_2 = \left(\sum_{i=1}^{n} (x_i)^2\right)^{\frac{1}{2}}$$



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Mathematical Formulation

Mathematical Program : A mathematical formulation of an optimization problem:

$$\text{Minimize } f(x) \text{ subject to } x \in S$$

Essential Components of a Mathematical program:

x: variables or parameters, f : objective function, S: feasible region

What is a solution of this Mathematical Program?

$$x^* \in S \text{ such that } f(x^*) \leq f(x) \ \forall \ x \in S$$

x^*: solution, $f(x^*)$: optimal objective function value

x^* may not be unique and may not even exist.

$$\text{Maximize } f(x) \equiv -\text{ Minimize } -f(x)$$

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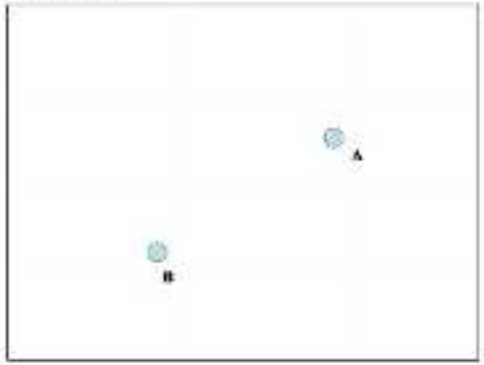
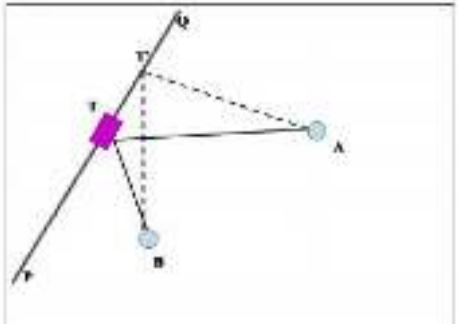
Examples

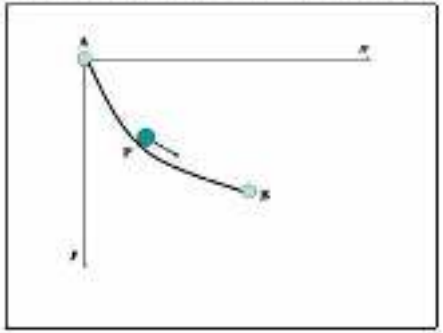
- Find the *shortest* path between the two points A and B in a horizontal plane



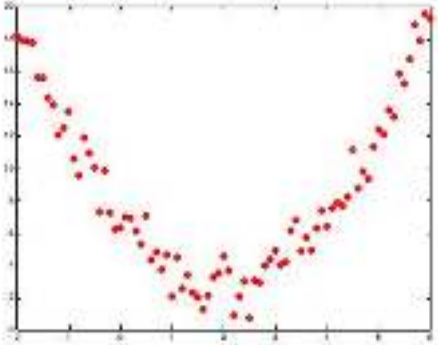
- Bus Terminus Location Problem: Find the location of the bus terminus T on the road segment PQ such that the lengths of the roads linking T with the two cities A and B is *minimum*.



- Given two points A and B in a vertical plane, find a path APB which an object must follow, so that starting from A, it reaches B in the *shortest* time under its own gravity.



- Data Fitting Problem: From a family of potential models, find a model that "best" fits the observed data.



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Mathematical Optimization Process

Typical steps for Solving Mathematical Optimization Problems

- Problem formulation
- Checking the existence of a solution
- Solving the optimization problem, if a solution exists
- Solution analysis
- Algorithm analysis

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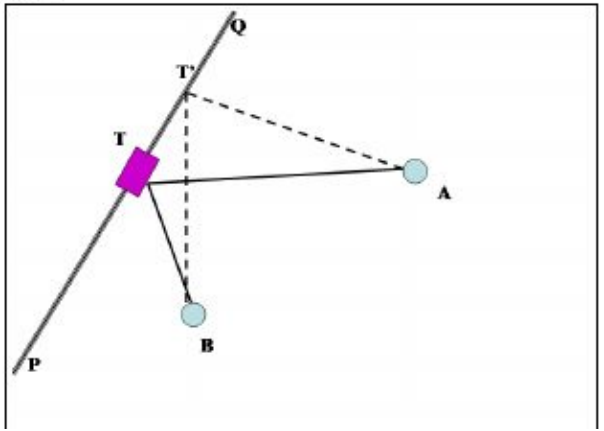
Problem formulation

- Bus Terminus Location Problem: Find the location of the bus terminus T on the road segment PQ such that the lengths of the roads linking T with the two cities A and B is *minimum*.



- Coordinates of A and B:
  $x_A = (x_{A1}, x_{A2})$ and

  $x_B = (x_{B1}, x_{B2})$

- Equation of line PQ:
  $ax_1 + bx_2 + C = 0$

- Use Euclidean distance
  $x_T = x_{T1}, x_{T2}$ (*variables*)

- The objective is to minimize
  $d(x_A, x_T) + d(x_B, x_T)$

- T lies on PQ (constraint)
  $\min = d(x_A, x_T) + d(x_B, x_T)$
  x_{T1}, x_{T2}

  s.t. $ax_{T1} + bx_{T2} + c = 0$

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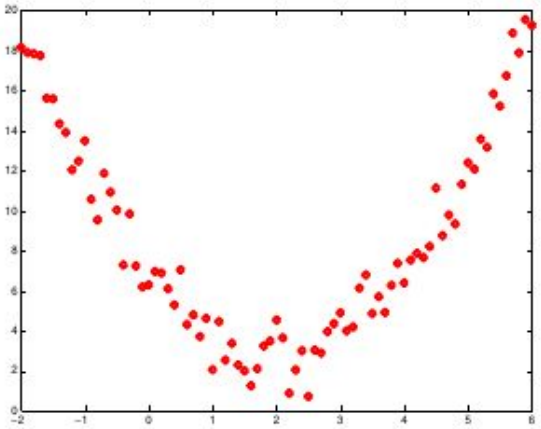
Problem formulation

- Data Fitting Problem: From a family of potential models, find a model that "best" fits the observed data.



- Given: $\{x_i, y_i\}_{i=1}^{n}$, n data points
- Given most probable model type,
  $f(x) = ax^2 + bx + c$
  a, b, c : variables
- Measure of Misfit: $(y-f(x))^2$
- The objective is to minimize
  $(y_i - (ax_i^2 + bx_i + c))^2$
- No constraint
  $\min_{a,b,c} \sum (y_i - (ax_i^2 + bx_i + c)^2)$

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Mathematical Optimization Process

Typical steps for Solving Mathematical Optimization Problems

- Problem formulation
- Checking the existence of a solution
- Solving the optimization problem, if a solution exists
  - Graphical method
  - Analytical method
  - Numerical method
- Solution analysis
- Algorithm analysis

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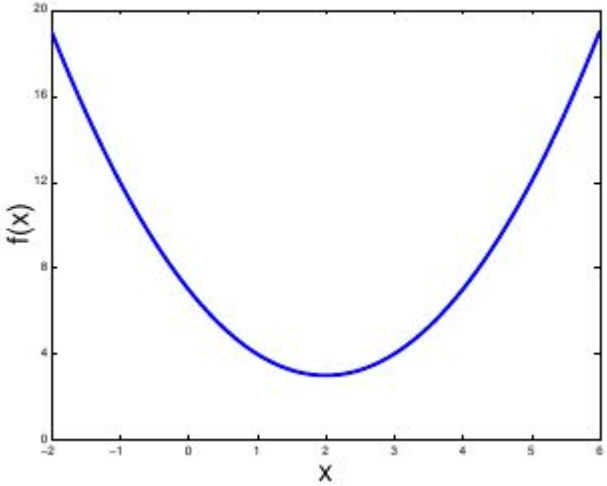
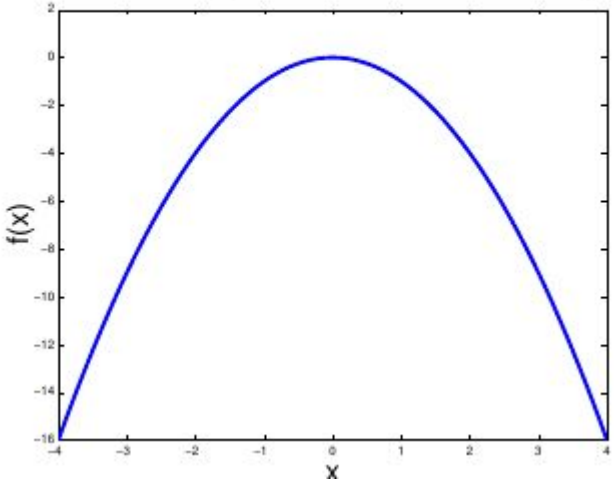
Graphical Method (example)

- $f(x) = (x-2)^2 + 3$
- Minimum at $x^* = 2$, minimum function value: $f(x^*) = 3$



- $f(x) = -x^2$
- Maximum at $x^* = 0$, maximum function value: $f(x^*) = 0$



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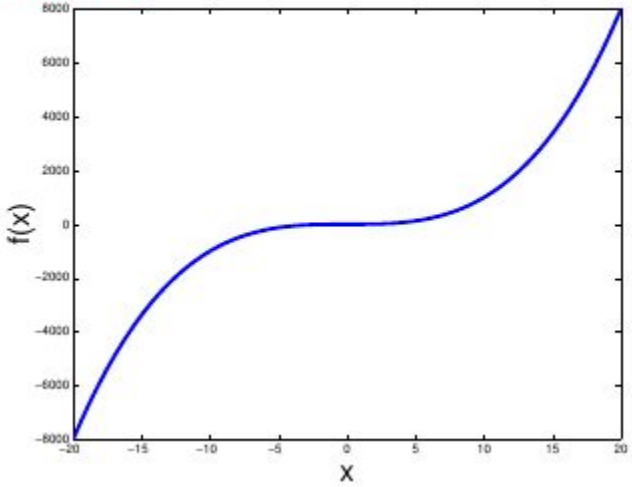
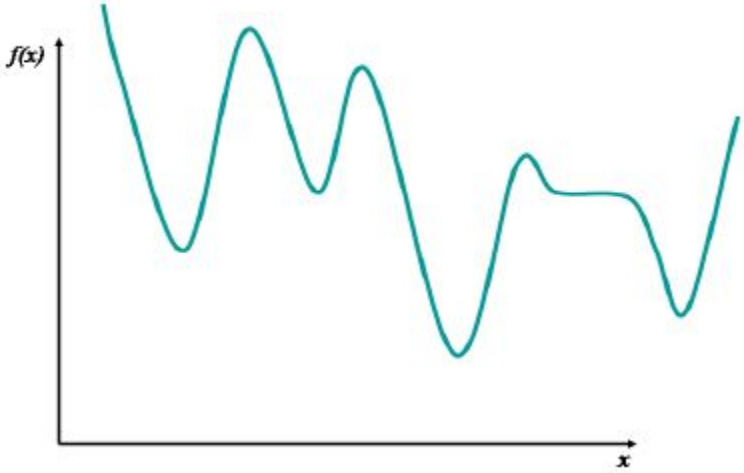
Graphical Method (example)

- $f(x) = x^3$
- Saddle Point at $x^* = 0$



- A typical nonlinear function



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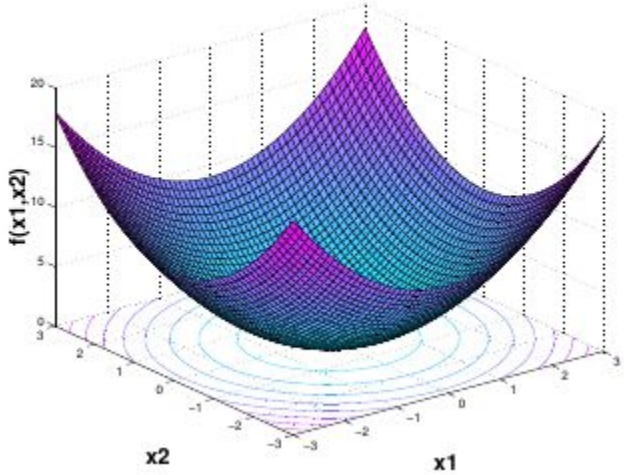
Graphical Method (example)

- $f(x_1, x_2) = x_1^2 + x_2^2$
- Minimum at $x_1^* = 0$, $x_2^* = 0$; $f(x_1^*, x_2^*) = 0$



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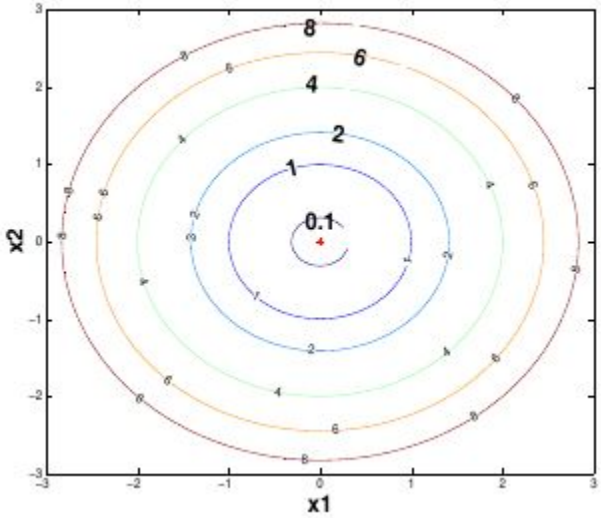

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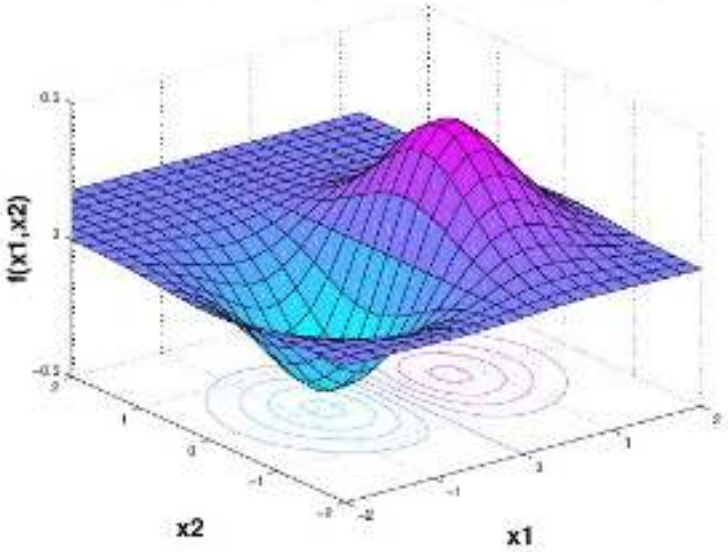
Graphical Method (example)

- $f(x_1, x_2) = x_1 \exp(-x_1^2 - x_2^2)$
- Minimum at $(-1/\sqrt{2}, 0)$, maximum at $(1/\sqrt{2}, 0)$



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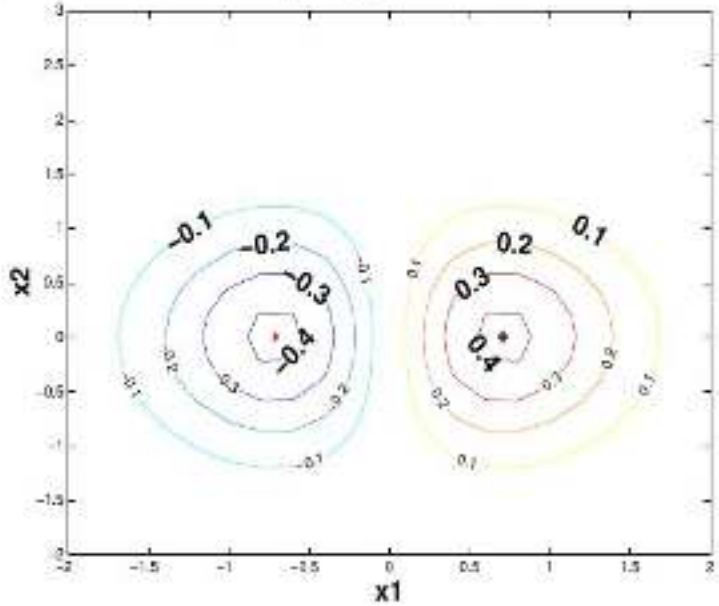

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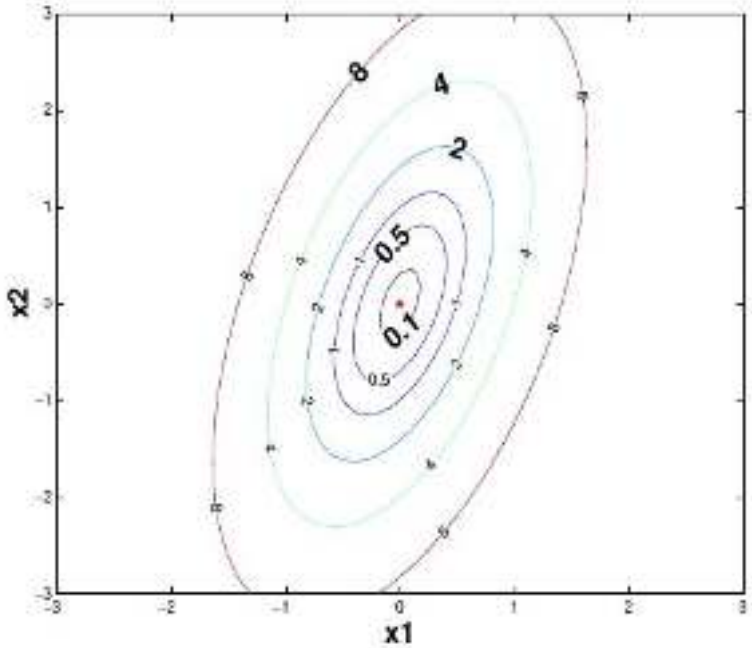
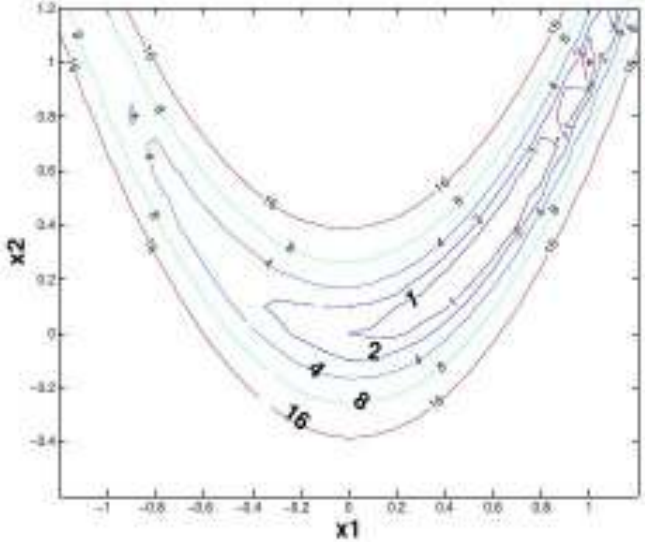
Graphical Method (example)

- $f(x_1, x_2) = f(x_1, x_2) = 4x_1^2 + x_2^2 - 2x_1x_2$



- Rosenbrock function: $f(x_1, x_2) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$
- Minimum at $(1, 1)$



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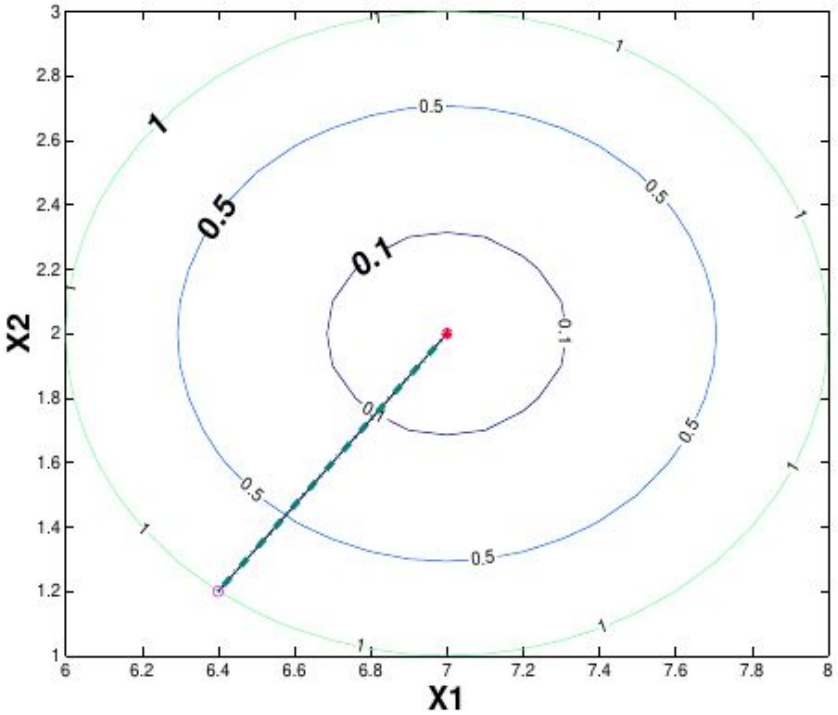
Numerical Method (example of iterates of an optimization algorithm)

- $f(x_1, x_2) = (x_1 - 7)^2 + (x_2 - 2)^2$
- Initial Point: $(6.4, 1.2)$, Minimum at $(7, 2)$



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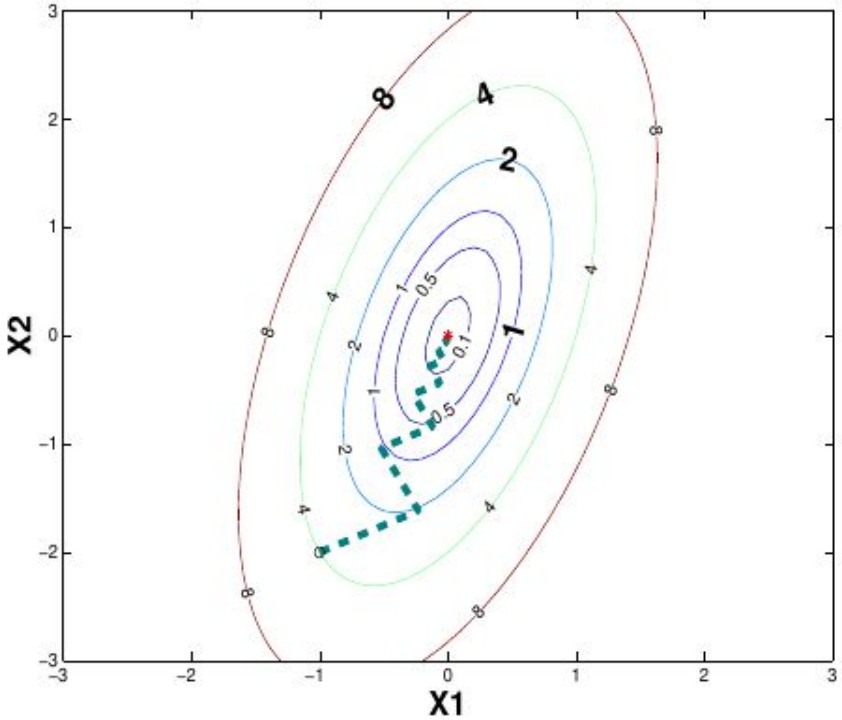
Numerical Method (example of iterates of an optimization algorithm)

- $f(x_1, x_2) = 4x_1^2 + x_2^2 - 2x_1x_2$
- Initial Point: $(-1, -2)$, Minimum at $(0, 0)$



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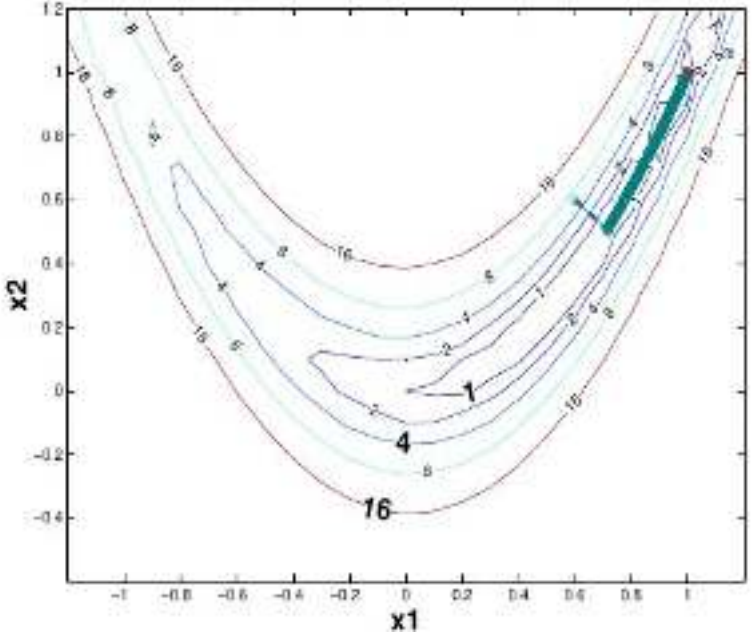
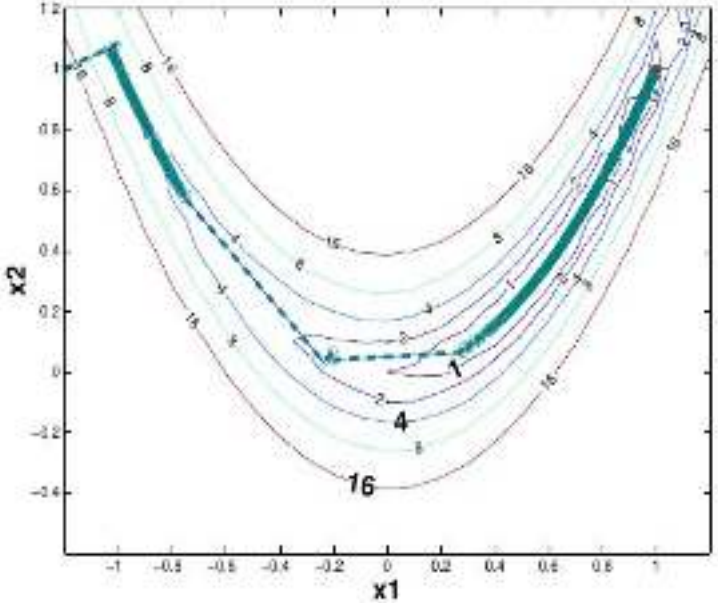
Numerical Method (example of iterates of an optimization algorithm)

- $f(x_1, x_2) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$
- Initial Point: $(0.6, 0.6)$, Minimum at $(1, 1)$



- $f(x_1, x_2) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$
- Initial Point: $(-2, 1)$, Minimum at $(1, 1)$



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Global minimum

Let $X \subseteq \mathbb{R}^n$ and $f : X \to \mathbb{R}$
Consider the problem,

Constrained optimization problem

$$\begin{aligned} \min_{\boldsymbol{x}} \quad & f(\boldsymbol{x}) \\ \text{s.t.} \quad & \boldsymbol{x} \in X \end{aligned}$$

Definition

$\boldsymbol{x}^* \in X$ is is said to be a *global minimum* of f over X if
$f(\boldsymbol{x}^*) \leq f(\boldsymbol{x}) \quad \forall\, \boldsymbol{x} \in X.$

Question: Under what conditions on f and X does the function f attain its maximum and/or minimum in the set X?

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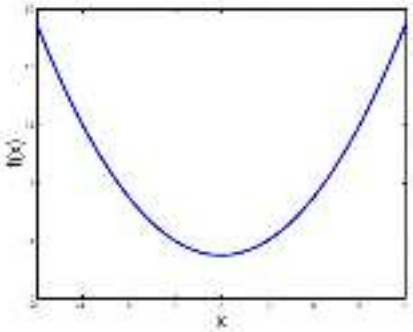
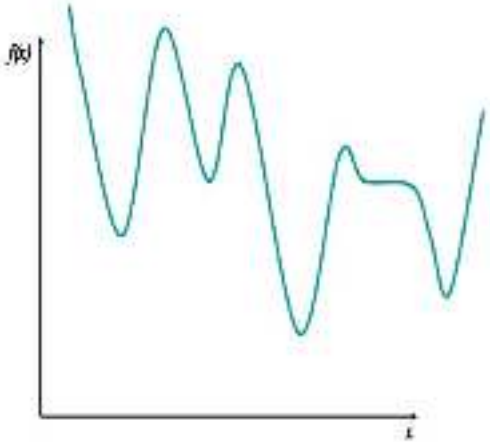
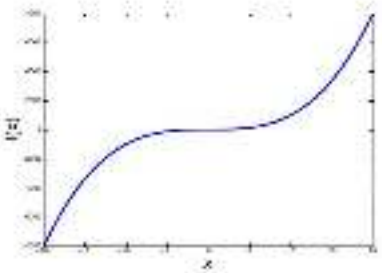
Global minimum

Concept of compact set: Closed and bounded.





- $X = \mathbb{R}, f : X \to \mathbb{R}$ defined as $f(x) = x^3$.



f attains neither a minimum nor a maximum on X

Note: X is closed, but not bounded; that is, X is not a compact set

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Global minimum

Concept of compact set: Closed and bounded.

Theorem

Let $X \subset \mathbb{R}^n$ be a nonempty compact set and $f : X \to \mathbb{R}$ be a continuous function on X. Then, f attains a maximum and a minimum on X; that is, there exist $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ in X such that

$$f(\boldsymbol{x}_1) \geq f(\boldsymbol{x}) \geq f(\boldsymbol{x}_2) \quad \forall \boldsymbol{x} \in X.$$

Note: Weierstrass' Theorem provides only *sufficient* conditions for the existence of optima.

Reference :NPTEL Course on Numerical Optimization by Prof. Shirish Shevade, IISc Bangalore.

Global minimum

Let $X \subseteq \mathbb{R}^n$ and $f : X \to \mathbb{R}$
Consider the problem,

Constrained optimization problem

$$\begin{aligned} \min_{x} \quad & f(x) \\ \text{s.t.} \quad & x \in X \end{aligned}$$

Definition

$\boldsymbol{x}^* \in X$ is is said to be a *global minimum* of f over X if
$f(\boldsymbol{x}^*) \leq f(\boldsymbol{x}) \quad \forall\, \boldsymbol{x} \in X.$

- Global minimum is difficult to find or characterize for a general nonlinear function

Reference :NPTEL Course on Numerical Optimization by Prof. Shirish Shevade, IISc Bangalore.

Local minimum

Let $X \subseteq \mathbb{R}^n$ and $f : X \to \mathbb{R}$
Consider the problem,

Constrained optimization problem

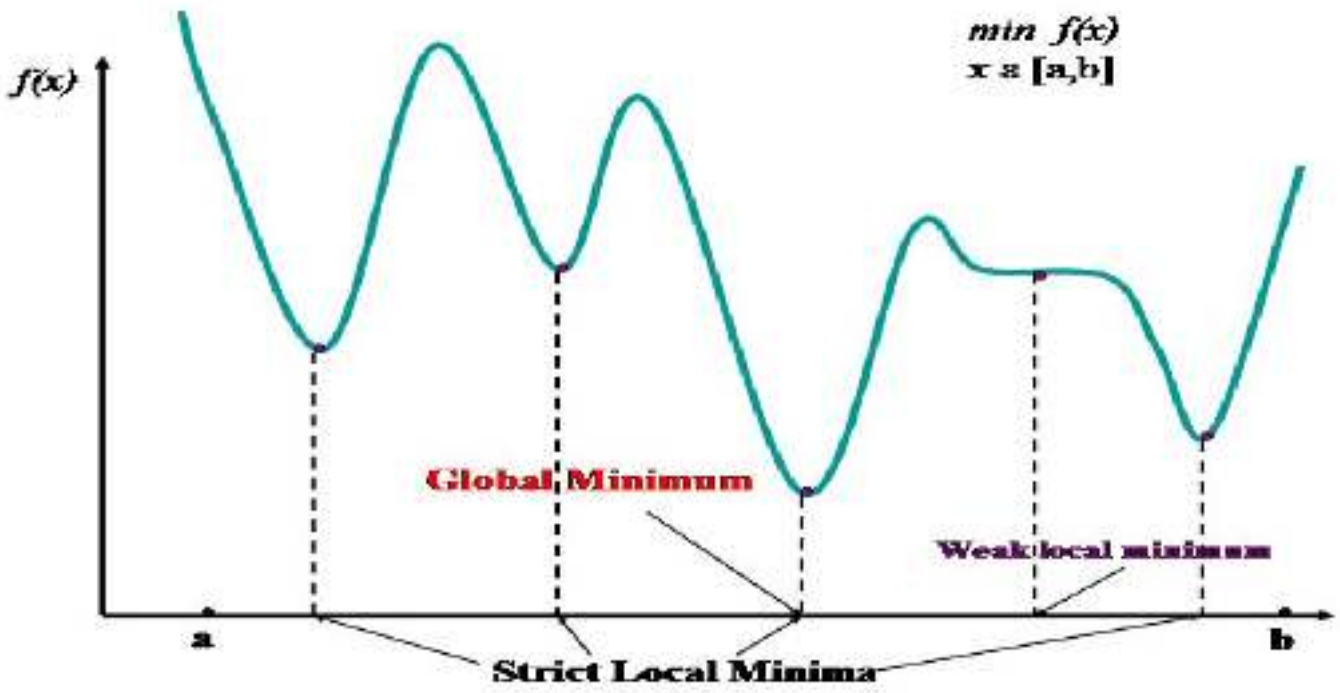
$$\begin{aligned} \min_{x} \quad & f(x) \\ \text{s.t.} \quad & x \in X \end{aligned}$$

Definition

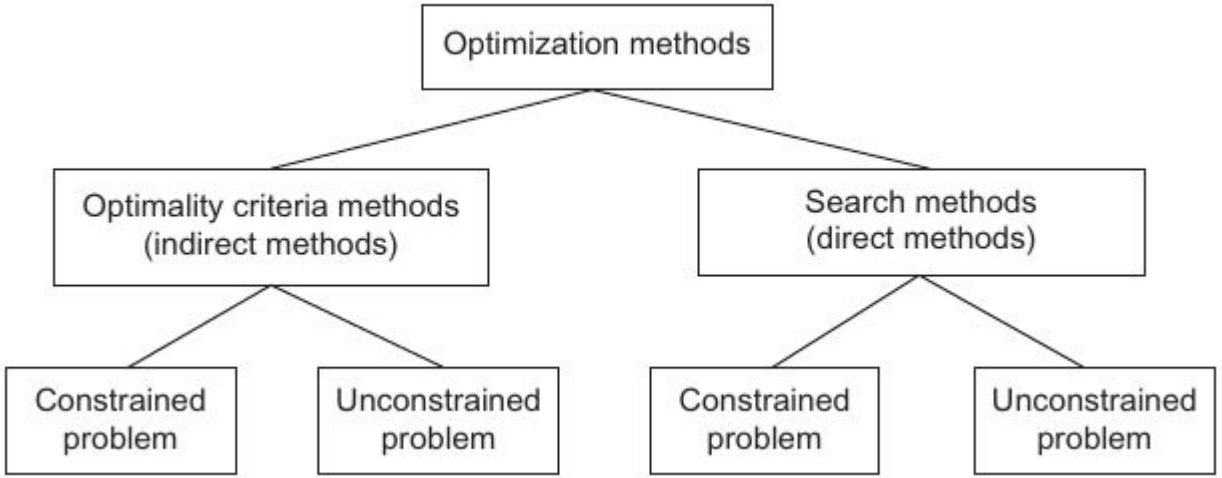
$\boldsymbol{x}^* \in X$ is is said to be a *local minimum* of f if there is a $\delta > 0$ such that $f(\boldsymbol{x}^*) \leq f(\boldsymbol{x}) \quad \forall\, \boldsymbol{x} \in X \cap B(\boldsymbol{x}^*, \delta)$.

Reference :NPTEL Course on Numerical Optimization by Prof. Shirish Shevade, IISc Bangalore.

Different type of minimas



Reference :NPTEL Course on Numerical Optimization by Prof. Shirish Shevade, IISc Bangalore.



- Optimality criteria are the requirements which the function should satisfy at the optimal point.
- Search methods start with an initial guess and iterate through the possible states to reach an optimal solutions.

Here we discuss about Minimization problems (Min f(**x**)), you can formulate a maximization problem as Min (-f(**x**)).

Reference :NPTEL Course on Numerical Optimization by Prof. Shirish Shevade, IISc Bangalore.

Definition of global and local minima

Cost function : $f(\mathbf{x})$
Design variable vector : $\mathbf{X}$

Problem definition**:** Find the design variable vector $\mathbf{x}$ to minimize $f(\mathbf{x})$ subject to

- Equality constraints: $h_j(\mathbf{x}) = 0$; j= 1 to p
- Inequality constraints : $g_i(\mathbf{x}) <= 0$; i= 1 to m

Feasible set *S* : It is basically the set which contains a collection of feasible design for the design problem under consideration. It is also known as the constraint set or the feasible region or the feasible design space.

$$S = \{\mathbf{x} \mid h_j(\mathbf{x}) = 0;\ j= 1 \text{ to } p;\ \ g_i(\mathbf{x}) <= 0;\ i= 1 \text{ to } m\}$$

The *optimization problem* is to find a point in the feasible design space that gives the minimum value of the cost function.

Reference :NPTEL Course on Numerical Optimization by Prof. Shirish Shevade, IISc Bangalore; Introduction to optimum design 3rd edn. by Jasbir S Arora, Academic Press.

Global (Absolute) Minimum

A function $f(\mathbf{x})$ of n variables has a global (absolute) minimum at $\mathbf{x}^*$ if the value of the function at $\mathbf{x}^*$ is less than or equal to the value of the function at any other point $\mathbf{x}$ in the feasible set S. That is,

$$f(\mathbf{x}^*) \leq f(\mathbf{x}) \tag{4.2}$$

for all $\mathbf{x}$ in the feasible set S. If strict inequality holds for all $\mathbf{x}$ other than $\mathbf{x}^*$ in Eq. (4.2), then $\mathbf{x}^*$ is called a *strong (strict) global minimum*; otherwise, it is called a *weak global minimum*.

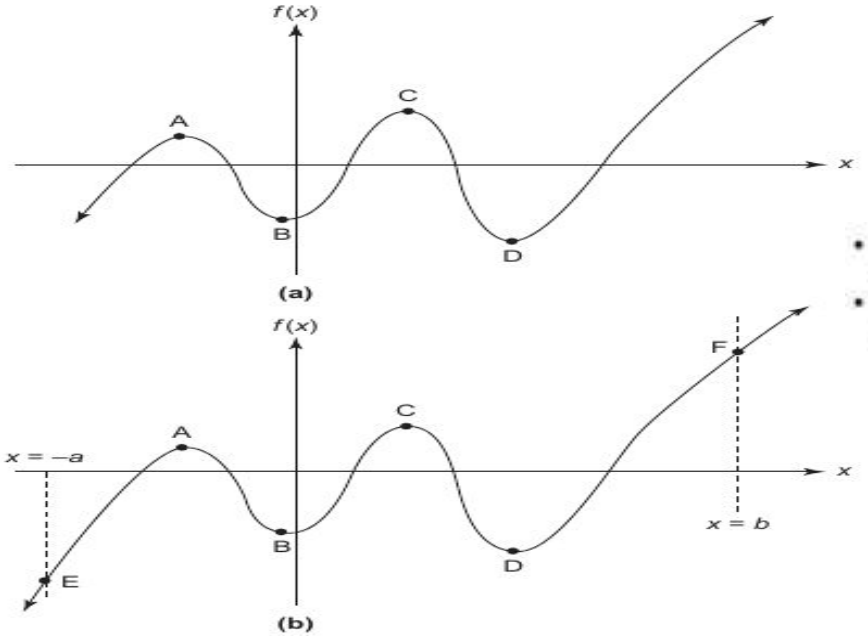
Local (Relative) Minimum

A function $f(\mathbf{x})$ of n variables has a local (relative) minimum at $\mathbf{x}^*$ if Inequality (4.2) holds for all $\mathbf{x}$ in a small *neighborhood* N (vicinity) of $\mathbf{x}^*$ in the feasible set S. If strict inequality holds, then $\mathbf{x}^*$ is called a *strong (strict) local minimum*; otherwise, it is called a *weak local minimum*.

Neighborhood N of point $\mathbf{x}^*$ is defined as the set of points

$$N = \{\mathbf{x} | \mathbf{x} \in S \text{ with } ||\mathbf{x} - \mathbf{x}^*|| < \delta\} \tag{4.3}$$

for some small $\delta > 0$. Geometrically, it is a small feasible region around point $\mathbf{x}^*$.



- A global minimum point is the one where there are no other feasible points with better cost function values.
- A local minimum point is the one where there are no other feasible points "in the vicinity" with better cost function values.

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An optimum design problem is formulated and transcribed into the standard form in terms of the variables x and y:

Minimize

$$f(x, y) = (x-4)^2 + (y-6)^2 \tag{a}$$

subject to

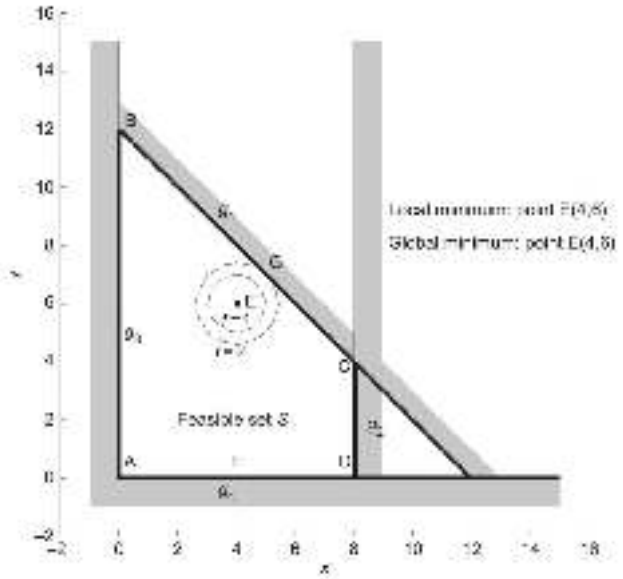
$$g_1 = x + y - 12 \le 0 \tag{b}$$

$$g_2 = x - 8 \le 0 \tag{c}$$

$$g_3 = -x \le 0 \ (x \ge 0) \tag{d}$$

$$g_4 = -y \le 0 \ (y \ge 0) \tag{e}$$

Find the local and global minima for the function $f(x,y)$ using the graphical method.



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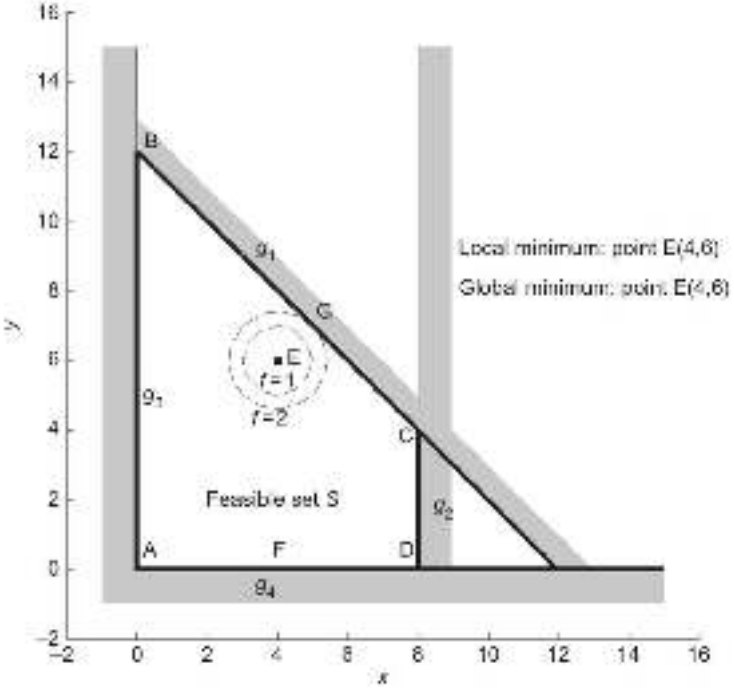
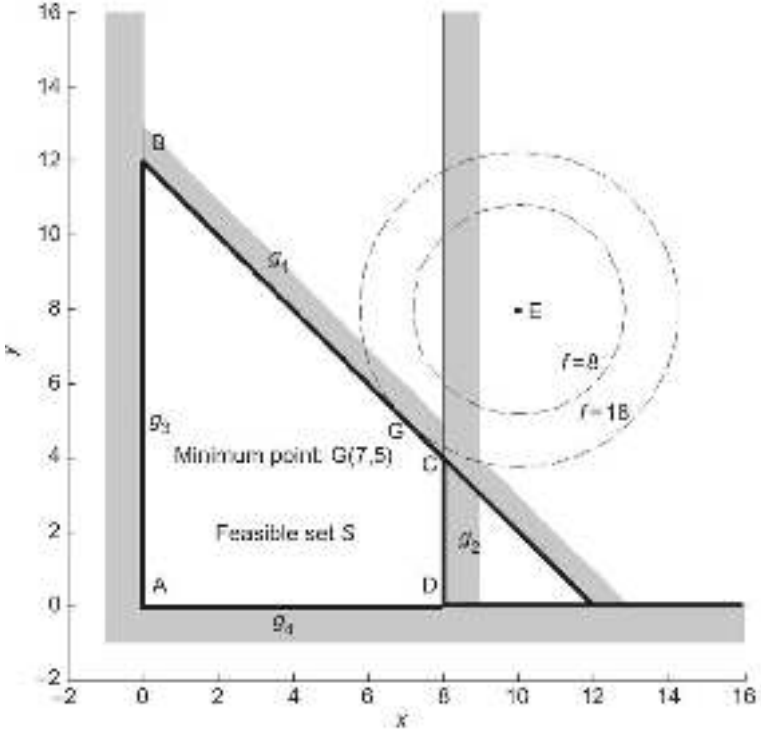


$$f(x,y) = (x-4)^2 + (y-6)^2$$



$$f(x,y) = (x-10)^2 + (y-8)^2$$

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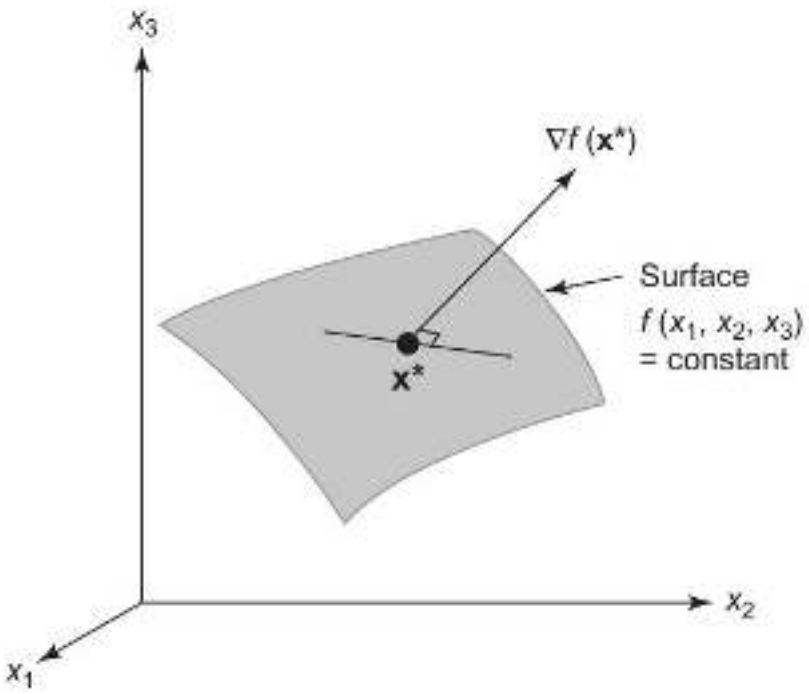
THEOREM 4.1

Weierstrass Theorem—Existence of a Global Minimum If $f(\mathbf{x})$ is continuous on a non-empty feasible set S that is closed and bounded, then $f(\mathbf{x})$ has a global minimum in S.

Note: When the conditions of the Weierstrass theorem are satisfied, the existence of a global optimum is guaranteed.
However, the theorem does not propose an “if-and-only-if” condition. It does not rule out the possibility of existence of a global minimum even when not satisfied.

e.g. Min $f(x) = x^2$ $-1 < x < 1$; since feasible set is not closed, conditions of the theorem are not met. However, function has a global minimum at $x=0$.

Review of some basic calculus concepts



Gradient vector for a function of "n" variables: $f(x) = f(x_1, x_2, x_3 \ldots\ldots x_n)$; Here **x*** represents the optimal point.

$$c_i = \frac{\partial f(\mathbf{x}^*)}{\partial x_i}; \quad i = 1 \text{ to } n$$

$$\mathbf{c} = \nabla f(\mathbf{x}^*) = \begin{bmatrix} \dfrac{\partial f(\mathbf{x}^*)}{\partial x_1} \\ \dfrac{\partial f(\mathbf{x}^*)}{\partial x_2} \\ \vdots \\ \dfrac{\partial f(\mathbf{x}^*)}{\partial x_n} \end{bmatrix} = \begin{bmatrix} \dfrac{\partial f(\mathbf{x}^*)}{\partial x_1} & \dfrac{\partial f(\mathbf{x}^*)}{\partial x_2} & \cdots & \dfrac{\partial f(\mathbf{x}^*)}{\partial x_n} \end{bmatrix}^T$$

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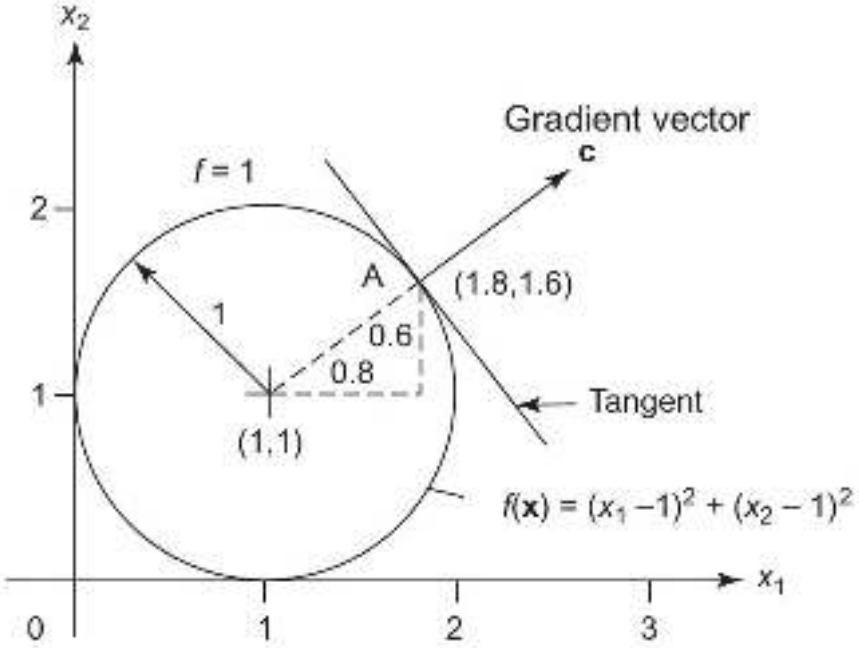
Review of some basic calculus concepts

Calculate the gradient vector for the function $f(\mathbf{x}) = (x_1 - 1)^2 + (x_2 - 1)^2$ at the point $\mathbf{x}^* = (1.8, 1.6)$.



$$\frac{\partial f}{\partial x_1}(1.8, 1.6) = 2(x_1 - 1) = 2(1.8 - 1) = 1.6$$

$$\frac{\partial f}{\partial x_2}(1.8, 1.6) = 2(x_2 - 1) = 2(1.6 - 1) = 1.2$$

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Review of some basic calculus concepts

The Hessian Matrix : Second order partial derivatives obtained by differentiating the gradient vector once again w.r.t **x**

$$\frac{\partial^2 f}{\partial \mathbf{x} \partial \mathbf{x}} = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

$$\mathbf{H} = \left[\frac{\partial^2 f}{\partial x_j \partial x_i} \right]; \quad i = 1 \text{ to } n, \; j = 1 \text{ to } n$$

The Hessian Matrix or the Hessian plays an important role in the sufficiency condition for optimality.

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Review of some basic calculus concepts

The Hessian Matrix : Second order partial derivatives obtained by differentiating the gradient vector once again w.r.t **x**

For the following function, calculate the gradient vector and the Hessian matrix at the point (1, 2):

$$f(\mathbf{x}) = x_1^3 + x_2^3 + 2x_1^2 + 3x_2^2 - x_1x_2 + 2x_1 + 4x_2 \tag{a}$$

Solution

The first partial derivatives of the function are given as

$$\frac{\partial f}{\partial x_1} = 3x_1^2 + 4x_1 - x_2 + 2; \quad \frac{\partial f}{\partial x_2} = 3x_2^2 + 6x_2 - x_1 + 4 \tag{b}$$

Substituting the point $x_1 = 1$, $x_2 = 2$, the gradient vector is given as $\mathbf{c} = (7, 27)$.

The second partial derivatives of the function are calculated as

$$\frac{\partial^2 f}{\partial x_1^2} = 6x_1 + 4; \quad \frac{\partial^2 f}{\partial x_1 \partial x_2} = -1; \quad \frac{\partial^2 f}{\partial x_2 \partial x_1} = -1; \quad \frac{\partial^2 f}{\partial x_2^2} = 6x_2 + 6. \tag{c}$$

Therefore, the Hessian matrix is given as

$$\mathbf{H}(\mathbf{x}) = \begin{bmatrix} 6x_1 + 4 & -1 \\ -1 & 6x_2 + 6 \end{bmatrix} \tag{d}$$

The Hessian matrix at the point (1, 2) is given as

$$\mathbf{H}(1, 2) = \begin{bmatrix} 10 & -1 \\ -1 & 18 \end{bmatrix} \tag{e}$$

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Review of some basic calculus concepts

The Taylor's series expansion : Taylor's expansion for a function of single variable f(x) about a point x* is given by:

$$f(x) = f(x^*) + \frac{df(x^*)}{dx}(x - x^*) + \frac{1}{2}\frac{d^2f(x^*)}{dx^2}(x - x^*)^2 + R$$

R is the remainder.
Let d= x-x*, then the equation can be written as follows:

$$f(x^* + d) = f(x^*) + \frac{df(x^*)}{dx}d + \frac{1}{2}\frac{d^2f(x^*)}{dx^2}d^2 + R$$

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Review of some basic calculus concepts

R is the remainder.
Let d= x-x*, then the equation can be written as follows:

$$f(x^* + d) = f(x^*) + \frac{df(x^*)}{dx} d + \frac{1}{2} \frac{d^2 f(x^*)}{dx^2} d^2 + R$$

Similarly, for a function of two variables, x_1 and x_2, the Taylor's series expansion can be written as follows:
Here, $d_1 = x_1 - x_1^*$ and $d2 = x_2 - x_2^*$:

$$f(x_1, x_2) = f(x_1^*, x_2^*) + \frac{\partial f}{\partial x_1} d_1 + \frac{\partial f}{\partial x_2} d_2 + \frac{1}{2} \left[\frac{\partial^2 f}{\partial x_1^2} d_1^2 + 2 \frac{\partial^2 f}{\partial x_1 \partial x_2} d_1 d_2 + \frac{\partial^2 f}{\partial x_2^2} d_2^2 \right]$$

All partial derivatives are calculated at (x_1^* , x_2^*)

In summation notation it can be written as:

$$f(x_1, x_2) = f(x_1^*, x_2^*) + \sum_{i=1}^{2} \frac{\partial f}{\partial x_i} d_i + \frac{1}{2} \sum_{i=1}^{2} \sum_{j=1}^{2} \frac{\partial^2 f}{\partial x_i \partial x_j} d_i d_j$$

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Review of some basic calculus concepts

All partial derivatives are calculated at (x_1^* , x_2^*)

In summation notation it can be written as:

$$f(x_1, x_2) = f(x_1^*, x_2^*) + \sum_{i=1}^{2} \frac{\partial f}{\partial x_i} d_i + \frac{1}{2} \sum_{i=1}^{2} \sum_{j=1}^{2} \frac{\partial^2 f}{\partial x_i \partial x_j} d_i d_j$$

In matrix notation for a vector **X,** Taylor's series expansion can be written as follows:

$$f(\mathbf{x}^* + \mathbf{d}) = f(\mathbf{x}^*) + \nabla f^T \mathbf{d} + \frac{1}{2} \mathbf{d}^T \mathbf{H} \mathbf{d} + R$$

Here, ∇f represents the gradient vector and **H** represents the Hessian matrix.

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Review of some basic calculus concepts

In matrix notation for a vector **X,** Taylor's series expansion can be written as follows:

$$f(\mathbf{x}^* + \mathbf{d}) = f(\mathbf{x}^*) + \nabla f^T \mathbf{d} + \frac{1}{2}\mathbf{d}^T \mathbf{H}\mathbf{d} + R$$

Here, ∇f represents the gradient vector and **H** represents the Hessian matrix.
If **x** = (x_1* , x_2*), **H** will be the 2 x 2 Hessian matrix.
For a function of 'n' variables, **x**, **x*** and ∇f are n-dimensional vectors and **H** is the n x n Hessian matrix.

$$\Delta f = f(\mathbf{x}) - f(\mathbf{x}^*)$$

$$\Delta f = \nabla f^T \mathbf{d} + \frac{1}{2}\mathbf{d}^T \mathbf{H}\mathbf{d} + R$$

For a first order change, only the first term in the above equation is sufficient.

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Q1

Approximate $f(x) = \cos x$ around the point $x^* = 0$.

Solution

Derivatives of the function $f(x)$ are given as

$$\frac{df}{dx} = -\sin x, \quad \frac{d^2f}{dx^2} = -\cos x \tag{a}$$

Therefore, using Eq. (4.9), the second-order Taylor's expansion for $\cos x$ at the point $x^* = 0$ is given as

$$\cos x \approx \cos 0 - \sin\ 0(x-0) + \frac{1}{2}(-\cos\ 0)(x-0)^2 = 1 - \frac{1}{2}x^2 \tag{b}$$

Q2

Obtain a second-order Taylor's expansion for the function $f(\mathbf{x}) = 3x_1^3x_2$ at the point $\mathbf{x}^* = (1, 1)$.

Q2

The gradient and Hessian of the function $f(\mathbf{x})$ at the point $\mathbf{x}^* = (1,1)$ using Eqs. (4.5) and (4.8) are

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \dfrac{\partial f}{\partial x_1} \\ \dfrac{\partial f}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 9x_1^2 x_2 \\ 3x_1^3 \end{bmatrix} = \begin{bmatrix} 9 \\ 3 \end{bmatrix}; \quad \mathbf{H} = \begin{bmatrix} 18x_1x_2 & 9x_1^2 \\ 9x_1^2 & 0 \end{bmatrix} = \begin{bmatrix} 18 & 9 \\ 9 & 0 \end{bmatrix} \tag{a}$$

Substituting these in the matrix form of Taylor's expression given in Eq. (4.12), and using $\mathbf{d} = \mathbf{x} - \mathbf{x}^*$, we obtain an approximation $\bar{f}(\mathbf{x})$ for $f(\mathbf{x})$ as

$$\bar{f}(\mathbf{x}) = 3 + \begin{bmatrix} 9 \\ 3 \end{bmatrix}^T \begin{bmatrix} (x_1 - 1) \\ (x_2 - 1) \end{bmatrix} + \frac{1}{2} \begin{bmatrix} (x_1 - 1) \\ (x_2 - 1) \end{bmatrix}^T \begin{bmatrix} 18 & 9 \\ 9 & 0 \end{bmatrix} \begin{bmatrix} (x_1 - 1) \\ (x_2 - 1) \end{bmatrix} \tag{b}$$

where $f(\mathbf{x}^*) = 3$ has been used. Simplifying the expression by expanding vector and matrix products, we obtain Taylor's expansion for $f(\mathbf{x})$ about the point (1, 1) as

$$\bar{f}(\mathbf{x}) = 9x_1^2 + 9x_1x_2 - 18x_1 - 6x_2 + 9 \tag{c}$$

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Q2

The gradient and Hessian of the function $f(\mathbf{x})$ at the point $\mathbf{x}^* = (1,1)$ using Eqs. (4.5) and (4.8) are

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \dfrac{\partial f}{\partial x_1} \\ \dfrac{\partial f}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 9x_1^2 x_2 \\ 3x_1^3 \end{bmatrix} = \begin{bmatrix} 9 \\ 3 \end{bmatrix}; \quad \mathbf{H} = \begin{bmatrix} 18x_1x_2 & 9x_1^2 \\ 9x_1^2 & 0 \end{bmatrix} = \begin{bmatrix} 18 & 9 \\ 9 & 0 \end{bmatrix} \tag{a}$$

Substituting these in the matrix form of Taylor's expression given in Eq. (4.12), and using $\mathbf{d} = \mathbf{x} - \mathbf{x}^*$, we obtain an approximation $\bar{f}(\mathbf{x})$ for $f(\mathbf{x})$ as

$$\bar{f}(\mathbf{x}) = 3 + \begin{bmatrix} 9 \\ 3 \end{bmatrix}^T \begin{bmatrix} (x_1 - 1) \\ (x_2 - 1) \end{bmatrix} + \frac{1}{2} \begin{bmatrix} (x_1 - 1) \\ (x_2 - 1) \end{bmatrix}^T \begin{bmatrix} 18 & 9 \\ 9 & 0 \end{bmatrix} \begin{bmatrix} (x_1 - 1) \\ (x_2 - 1) \end{bmatrix} \tag{b}$$

where $f(\mathbf{x}^*) = 3$ has been used. Simplifying the expression by expanding vector and matrix products, we obtain Taylor's expansion for $f(\mathbf{x})$ about the point (1, 1) as

$$\bar{f}(\mathbf{x}) = 9x_1^2 + 9x_1x_2 - 18x_1 - 6x_2 + 9 \tag{c}$$

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Q2

This expression is a second-order approximation of the function $3x_1^3x_2$ about the point $\mathbf{x}^* = (1, 1)$. That is, in a small neighborhood of $\mathbf{x}^*$, the expression will give almost the same value as the original function $f(\mathbf{x})$. To see how accurately $\bar{f}(\mathbf{x})$ approximates $f(\mathbf{x})$, we evaluate these functions for a 30 percent change in the given point (1, 1), that is, at the point (1.3, 1.3) as $\bar{f}(\mathbf{x}) = 8.2200$ and $f(\mathbf{x}) = 8.5683$. Therefore, the approximate function underestimates the original function by only 4 percent. This is quite a reasonable approximation for many practical applications.

Q3

Obtain a linear Taylor's expansion for the function

$$f(\mathbf{x}) = x_1^2 + x_2^2 - 4x_1 - 2x_2 + 4 \qquad \text{(a)}$$

at the point $\mathbf{x}^* = (1, 2)$. Compare the approximate function with the original function in a neighborhood of the point (1, 2).

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Q3

Obtain a linear Taylor's expansion for the function

$$f(\mathbf{x}) = x_1^2 + x_2^2 - 4x_1 - 2x_2 + 4 \tag{a}$$

at the point $\mathbf{x}^* = (1, 2)$. Compare the approximate function with the original function in a neighborhood of the point (1, 2).

The gradient of the function at the point (1, 2) is given as

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \dfrac{\partial f}{\partial x_1} \\ \dfrac{\partial f}{\partial x_2} \end{bmatrix} = \begin{bmatrix} (2x_1 - 4) \\ (2x_2 - 2) \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \end{bmatrix} \tag{b}$$

Since $f(1, 2) = 1$, Eq. (4.13) gives a linear Taylor's approximation for $f(\mathbf{x})$ as

$$\bar{f}(\mathbf{x}) = 1 + [-2\ 2] \begin{bmatrix} (x_1 - 1) \\ (x_2 - 2) \end{bmatrix} = -2x_1 + 2x_2 - 1 \tag{c}$$

To see how accurately $\bar{f}(\mathbf{x})$ approximates the original $f(\mathbf{x})$ in the neighborhood of (1, 2), we calculate the functions at the point (1.1, 2.2), a 10 percent change in the point as $\bar{f}(\mathbf{x}) = 1.20$ and $f(\mathbf{x}) = 1.25$. We see that the approximate function underestimates the real function by 4 percent. An error of this magnitude is quite acceptable in many applications. Note, however, that the errors will be different for different functions and can be larger for highly nonlinear functions.

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Quadratic Form : A special non-linear function having only second order terms

$$F(\mathbf{x}) = x_1^2 + 2x_2^2 + 3x_3^2 + 2x_1x_2 + 2x_2x_3 + 2x_3x_1$$

Quadratic Form : Matrix form; **x** is a n-dimensional vector and **P** a n x n matrix.

$$F(\mathbf{x}) = \mathbf{x}^{\mathbf{T}}\mathbf{P}\mathbf{x}$$

Here **P** is known as the matrix of the quadratic form F(**x**).

There are many matrices associated with the given quadratic form; in fact, there are infinite such matrices. All of the matrices are asymmetric except one. The symmetric matrix **A** associated with the quadratic form can be obtained from any asymmetric matrix **P** as

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$$\mathbf{A} = \frac{1}{2}(\mathbf{P} + \mathbf{P}^T) \text{ or } a_{ij} = \frac{1}{2}(p_{ij} + p_{ji}), \quad i, j = 1 \text{ to } n$$

Here **P** is known as the matrix of the quadratic form F(**x**), **A** is the symmetric matrix, it is useful for determining the nature of the quadratic form.

Identify a matrix associated with the quadratic form:

$$F(x_1, x_2, x_3) = 2x_1^2 + 2x_1x_2 + 4x_1x_3 - 6x_2^2 - 4x_2x_3 + 5x_3^2$$

Writing F in the matrix form ($F(\mathbf{x}) = \mathbf{x}^{\mathbf{T}}\mathbf{P}\mathbf{x}$), we obtain

$$F(\mathbf{x}) = [x_1 \quad x_2 \quad x_3] \begin{bmatrix} 2 & 2 & 4 \\ 0 & -6 & -4 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

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$$\mathbf{P} = \begin{bmatrix} 2 & 0.5 & 1 \\ 1.5 & -6 & -6 \\ 3 & 2 & 5 \end{bmatrix}; \quad \mathbf{P} = \begin{bmatrix} 2 & 4 & 5 \\ -2 & -6 & 4 \\ -1 & -8 & 5 \end{bmatrix}$$

Here **P** is known as the matrix of the quadratic form F(**x**), **A** is the symmetric matrix, it is useful for determining the nature of the quadratic form.

Eigenvalue Check for the Form of a Matrix Let λ_i, $i = 1$ to n be the eigenvalues of a symmetric $n \times n$ matrix $\mathbf{A}$ associated with the quadratic form $F(\mathbf{x}) = \mathbf{x}^T\mathbf{A}\mathbf{x}$ (since $\mathbf{A}$ is symmetric, all eigenvalues are real). The following results can be stated regarding the quadratic form $F(\mathbf{x})$ or the matrix $\mathbf{A}$:

1. $F(\mathbf{x})$ is *positive definite if and only if* all eigenvalues of $\mathbf{A}$ are strictly positive; i.e., $\lambda_i > 0$, $i = 1$ to n.
2. $F(\mathbf{x})$ is *positive semidefinite if and only if* all eigenvalues of $\mathbf{A}$ are non-negative; i.e., $\lambda_i \geq 0$, $i = 1$ to n (note that at least one eigenvalue must be zero for it to be called positive semidefinite).
3. $F(\mathbf{x})$ is *negative definite if and only if* all eigenvalues of $\mathbf{A}$ are strictly negative; i.e., $\lambda_i < 0$, $i = 1$ to n.
4. $F(\mathbf{x})$ is *negative semidefinite if and only if* all eigenvalues of $\mathbf{A}$ are nonpositive; i.e., $\lambda_i \leq 0$, $i = 1$ to n (note that at least one eigenvalue must be zero for it to be called negative semidefinite).
5. $F(\mathbf{x})$ is *indefinite if* some $\lambda_i < 0$ and some other $\lambda_j > 0$.

Necessary and sufficient conditions

Assumption: It is assumed that one is already at the optimum point (x*) and then study the behaviour of the objective function and its derivative at that point.

Necessary conditions: Must be satisfied at the optimum point. If not satisfied, the point **can not** be an optimum point.

Note:

- Satisfaction of necessary conditions does not guarantee optimality of the point.
- This means that there can be non-optimum points which satisfy the necessary conditions.
- This also means that, number of points satisfying necessary conditions can be more than the number of optimal points.
- Points satisfying the necessary conditions are called the **candidate optimum points.**
- Therefore, we must perform further tests to distinguish between optimum and non optimum points, both satisfying the necessary conditions.

Necessary and sufficient conditions

Assumption: It is assumed that one is already at the optimum point (x*) and then study the behaviour of the objective function and its derivative at that point.

Sufficient conditions: If a candidate optimum point satisfies the sufficient condition, then it is indeed an optimum point.

Note:

- If the sufficient condition is not satisfied, or cannot be used, we may not be able to conclude that the candidate design point is optimum or not.

- Further analysis of the problem or higher order conditions may be required to conclude definitely about the nature of optimality of the candidate point.

Necessary and sufficient conditions

Summary:

- Optimum points **MUST** satisfy the necessary conditions. Points that **do not satisfy them cannot be optimum**.
- **Non-optimum points** may also **satisfy necessary conditions**.
- A candidate point (i.e. a point which satisfies the necessary conditions) **satisfying a sufficient condition is indeed optimum.**
- If sufficiency condition cannot be used or it is not satisfied, **one may not be able to draw any conclusions** about the optimality of the candidate point.

Necessary and sufficient conditions

First order necessary condition:

$$f'(x^*) = 0$$

For a function of single variable (x)

$$\nabla f(\mathbf{x}^*) = \mathbf{0}$$

For a function of multiple variable **(x)**

Sufficient condition (local minima):

$$f''(x^*) > 0$$

For a function of single variable (x)

$$\mathbf{d}^T \mathbf{H}(\mathbf{x}^*)\mathbf{d} > 0$$

For a function of multiple variable **(x)**

Second order necessary condition:

$$f''(x^*) \geq 0$$

For a function of single variable (x); equal to zero represent the necessary condition

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Necessary and sufficient conditions

What if f'(x*) =0 and f''(x*) =0….??

Necessary condition: f'''(x*) = 0

Sufficient condition : $f^{IV}(x^*)>0$

In general,

The lowest non-zero derivative must be even ordered for stationary points (necessary condition)
And should be positive for local minima (sufficient condition)

Find the local minima for the function

$$f(x) = x^2 - 4x + 4$$

Find the minimal for the function

$$f(x) = x^4$$

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Function of one variable minimize $f(x)$	Function of several variable minimize $f(\mathbf{x})$
First-order necessary condition: $f' = 0$. Any point satisfying this condition is called a stationary point; it can be a local maximum, local minimum, or neither of the two (inflection point)	*First-order necessary condition*: $\nabla f = \mathbf{0}$. Any point satisfying this condition is called a stationary point; it can be a local minimum, local maximum, or neither of the two (inflection point)
Second-order necessary condition for a local minimum: $f'' \geq 0$	*Second-order necessary condition* for a local minimum: $\mathbf{H}$ must be at least positive semidefinite
Second-order necessary condition for a local maximum: $f'' \leq 0$	*Second-order necessary condition* for a local maximum: $\mathbf{H}$ must be at least negative semidefinite
Second-order sufficient condition for a local minimum: $f'' > 0$	*Second-order sufficient condition* for a local minimum: $\mathbf{H}$ must be positive definite
Second-order sufficient condition for a local maximum: $f'' < 0$	*Second-order sufficient condition* for a local maximum: $\mathbf{H}$ must be negative definite
Higher-order necessary conditions for a local minimum or local maximum: Calculate a higher-ordered derivative that is not 0; all odd-ordered derivatives below this one must be 0	
Higher-order sufficient condition for a local minimum Highest nonzero derivative must be even-ordered and positive	

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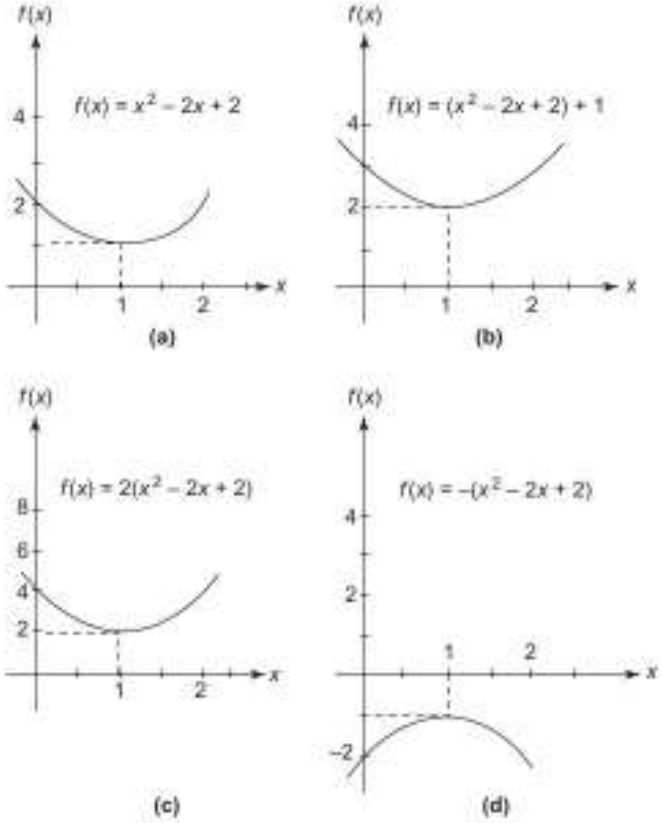

EFFECTS OF SCALING OR ADDING A CONSTANT TO THE COST FUNCTION

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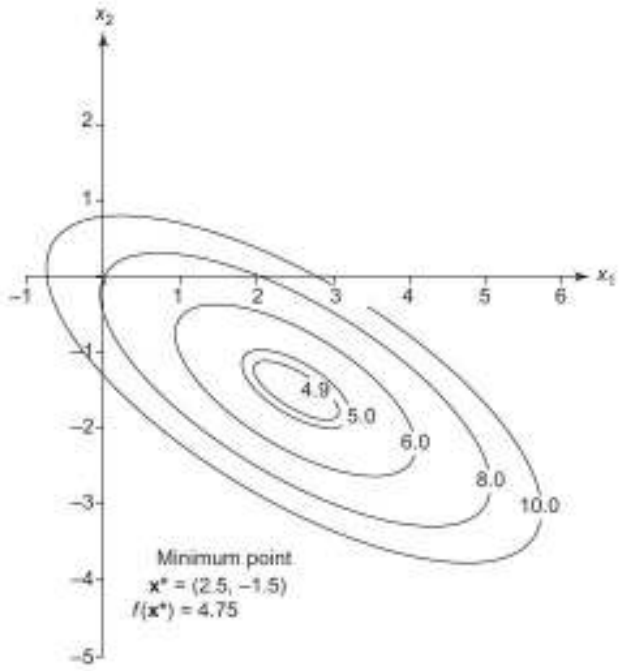
Find local minimum points for the function

$$f(\mathbf{x}) = x_1^2 + 2x_1x_2 + 2x_2^2 - 2x_1 + x_2 + 8$$



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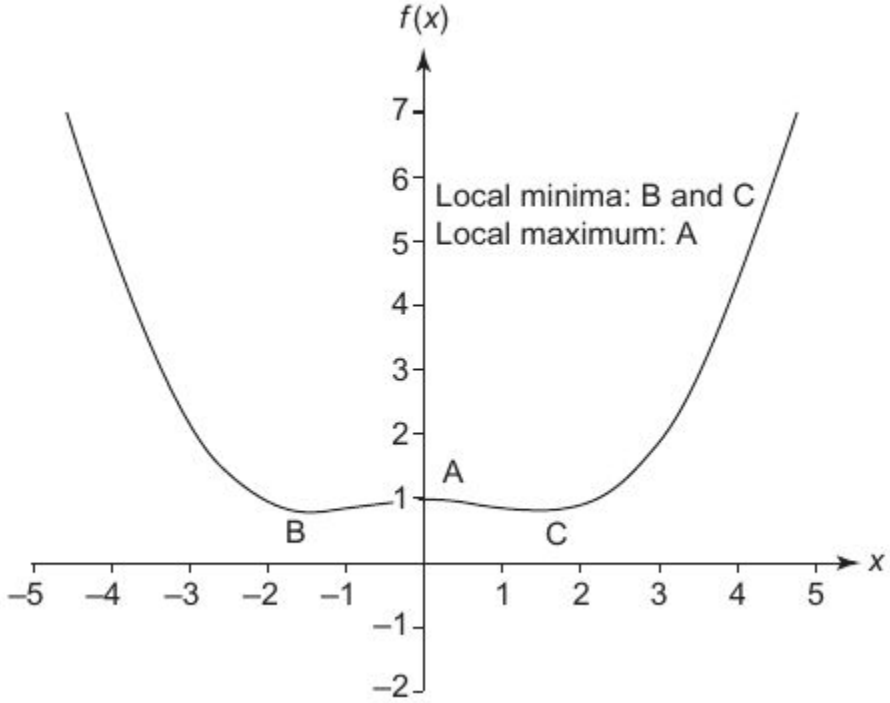
$$f(x) = \frac{1}{3}x^2 + \cos x$$



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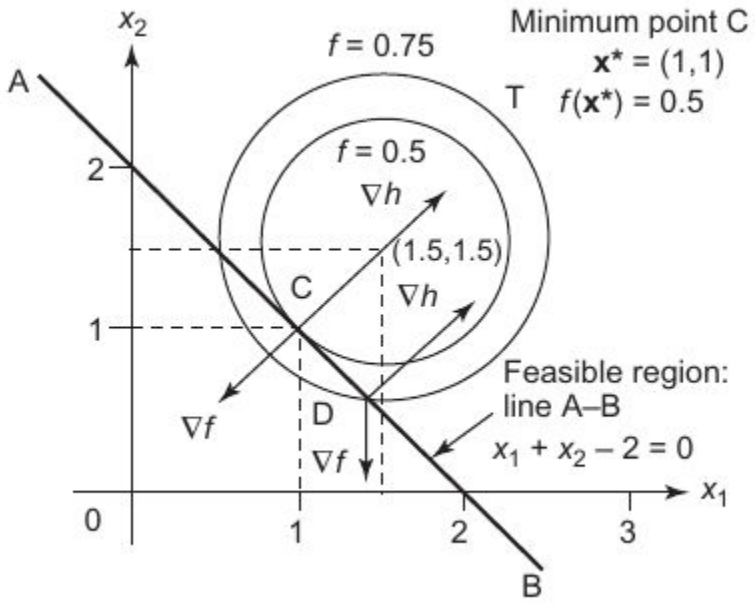
Lagrange multipliers and their geometrical meaning



Minimize

$$f(x_1, x_2) = (x_1 - 1.5)^2 + (x_2 - 1.5)^2$$

subject to an equality constraint:

$$h(x_1, x_2) = x_1 + x_2 - 2 = 0$$

$C = [1,1]^T$

$D = [0.4, 1.6]^T$

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CYLINDRICAL TANK DESIGN PROBLEM

A cylindrical storage tank needs to be fabricated. The tank is closed at both ends and is required to have volume V. The radius R and height H are selected as design variables. It is desired to design the tank having minimum surface area.

?

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